Triple helix to Schrödinger: a structural derivation chain from GToR axioms

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Abstract

We present a computational derivation chain that transports the helical 2-manifold of the General Theory of Reality (GToR) through the Dirac equation, the Klein–Gordon equation, and the non-relativistic Schrödinger equation. Each step is rendered as a theorem inside a `sympy` notebook; only one new technical claim is load-bearing: a Riccati slow-manifold reduction of Klein–Gordon that replaces the classical WKB ansatz. Three structural pieces of standard quantum mechanics are recovered from the geometry rather than postulated. First, the imaginary unit of the wavefunction is identified with the pseudoscalar $\sigma_1\sigma_2\sigma_3 = i\,I_2$ of $\mathrm{Cl}(3,0)$ in its spinor representation. Second, canonical quantization $\hat{p} = -i\hbar\partial_s$ is the spinor image of the screw-motion generator on the helix. Third, the first-order Dirac operator is the unique solution of the symbolic ansatz $\bar{D}D = (p{\cdot}p - m^2)I_4$ on the Clifford bundle defined by GToR's three rung directions. The dimensionless geometric prediction is $\alpha_\infty = 1/(14\pi^2)$, agreeing with the CODATA value of the fine-structure constant to 0.82% and to five significant figures after the atlas correction of GToR Cor 34. We are explicit about what the chain does *not* achieve: it does not derive the Lorentzian (1,3) signature from the seven GToR axioms, it does not "predict $\hbar$" in the post-2019 SI (where $\hbar$ is definitional), and the first-order restriction on differential operators is an acknowledged structural choice.

1 Introduction

The first-order operator of Dirac (1928), its second-order reduction to the Klein–Gordon equation (Klein, 1926; Gordon, 1926), and its non-relativistic limit recovering Schrödinger's equation (Schrödinger, 1926) together form the structural skeleton of single-particle quantum mechanics. The role of the Pauli matrices (Pauli, 1927) as the spinor generators of $\mathrm{Cl}(3,0)$ is classical. These results are usually introduced axiomatically in textbooks (Peskin and Schroeder, 1995; Weinberg, 1995), with the non-relativistic reduction carried out via a WKB ansatz or a Foldy–Wouthuysen decomposition (Foldy and Wouthuysen, 1950).

The General Theory of Reality (Yazdani, 2026b), hereafter GToR, develops the consequences of seven axioms describing a helical 2-manifold embedded in an expanding torus. Part I derives a chain of 56 corollaries, including a transfer matrix with exact trace $\mathrm{tr}(T_\ell) = (3/2)\cosh(\ell\sqrt{2}\ln 2)$, a closed-form fine-structure prediction $\alpha_\infty = 1/(14\pi^2)$ (Cor 31), and an atlas correction $\alpha(R) = \alpha_\infty + \delta(R)$ (Cor 34) that closes the gap to the CODATA value of α to five significant figures. The present paper is concerned with a complementary question: assuming the geometric setting of GToR, can the standard wave equations of relativistic and non-relativistic quantum mechanics be recovered as theorems?

Our answer is: yes for the structural skeleton, no for everything. We make explicit, line by line, where each fragment of the standard machinery enters as a theorem and where it is imported as a structural choice. The skeleton is then a *single Python notebook* (Yazdani, 2026a) of 39 cells, 18 of which execute symbolic calculations in `sympy` that verify each step.

The route is summarised in fig. 1.

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GToR axioms $\Rightarrow$ 3 orthogonal rung directions (Cor 21) $\Rightarrow$ Cl(3, 0) via Pauli matrices $\Rightarrow$ screw motion $\Rightarrow$ pseudoscalar $\Rightarrow$ canonical quantization
$\Rightarrow$ tensor with $\sigma_z \Rightarrow \gamma$-matrices / Clifford(1,3) $\Rightarrow$ first-order ansatz forces Dirac operator
$\Rightarrow$ $\bar{D}D = -(\Box + m^2)$ gives Klein–Gordon $\Rightarrow$ Riccati flow $\Rightarrow$ slow manifold $\Rightarrow$ Schrödinger.

Figure 1: Structural derivation chain. Bold steps are forced; italic links flag a structural choice (see Sec. 12).

Scope and honest claims. What is derived: the complex unit of QM as a geometric object, canonical quantization as a theorem on a helix, the unique first-order spinor operator squaring to the d'Alembertian plus mass, and the dimensionless prediction $\alpha_\infty = 1/(14\pi^2)$. What is not: the Lorentzian signature; the first-order restriction on differential operators; the selection of the positive-frequency Riccati branch; the rest-mass envelope subtraction $\phi := e^{imt}\psi$. These are flagged in section 12.

What is genuinely new. The structural recoveries above are repackagings of standard results (Clifford algebras, the Dirac operator, the Foldy–Wouthuysen non-relativistic projection (Foldy and Wouthuysen, 1950)) inside a single geometric setting. The technical novelty in the paper is the *Riccati slow-manifold reduction* of Klein–Gordon (section 10, with full proof in (Yazdani, 2026c)): we do not assume a WKB ansatz of the form $e^{-imt}\phi(x,t)$; instead we run a Riccati change of variables on the Klein–Gordon mode equation, identify the slow-manifold fixed points $u = \pm i\omega$, and integrate. The phase e^{-imt} then drops out of the integration as the leading slow-manifold term. The Schrödinger equation is the truncation of the resulting envelope equation at $O(k^2/m^2)$.

2 Geometric prerequisites from GToR

We collect the small set of GToR corollaries this paper depends on. For derivations and verification scripts see (Yazdani, 2026b).

- **Cor 21 (Toroidal embedding).** The expanding 2-manifold embeds in an ambient $\mathbb{R}^3$ with three orthogonal geometric directions: toroidal (e_1), poloidal (e_2), radial (e_3).
- **Cor 28 (Boundary metric).** $g(0) = 2$.
- **Cor 29 (Mass).** $m = \pi k^2/2$, with k the helix quantum.
- **Cor 30 (Decomposition).** $14\pi^2 = \sqrt{2}\, S\, V$, where S is the boundary surface and V the filled cell volume.
- **Cor 31 (Fine structure).** $\alpha_\infty = 1/(14\pi^2)$.
- **Cor 34 (Atlas correction).** $\alpha(R) = \alpha_\infty + \delta(R)$, matching CODATA α to five significant figures.

The starting point for the present paper is Cor 21: a Riemannian frame of three orthonormal vectors in an ambient $\mathbb{R}^3$.

3 Pauli matrices from the rung frame

We use the GToR rung frame to construct the three Pauli matrices, by a classical *representability* argument that we record here for completeness.

Proposition 3.1 (Plate 0)**.** *Three traceless Hermitian 2×2 involutions M_1, M_2, M_3 satisfying $\{M_i, M_j\} = 2\delta_{ij} I_2$ are in bijection with oriented orthonormal frames in $\mathbb{R}^3$.*

Proof. Parametrise a generic traceless Hermitian 2×2 matrix by $\boldsymbol{v} = (a, b, c) \in \mathbb{R}^3$,

$$M(\boldsymbol{v}) = \begin{pmatrix} c & a - ib \\ a + ib & -c \end{pmatrix}.$$

A direct calculation (verified symbolically in the accompanying notebook) gives $M(\boldsymbol{v})^2 = |\boldsymbol{v}|^2 I_2$ and $\{M(\boldsymbol{v}_1), M(\boldsymbol{v}_2)\} = 2(\boldsymbol{v}_1 \cdot \boldsymbol{v}_2)\, I_2$. Demanding the Clifford relation $\{M_i, M_j\} = 2\delta_{ij}\, I_2$ thus forces $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ to be orthonormal; choosing the standard frame yields the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

□

The point is that the rung frame of Cor 21 is exactly such a triple of oriented orthonormal vectors; the Pauli matrices are then the spinor representation of $\mathrm{Cl}(3, 0)$ (Cartan, 1938; Chevalley, 1954; Lawson and Michelsohn, 1989).

4 Screw motion and the bivector J

The helix propagates by *screw motion*: translation along an axis coupled to rotation in the transverse plane. Pick the axis as e_3; the transverse plane is spanned by e_1, e_2. Its oriented area element is the bivector

$$J := e_1 e_2 \in \mathrm{Cl}(3, 0).$$

In the spinor representation $e_i \mapsto \sigma_i$, bivectors $e_i e_j$ map to $\sigma_i \sigma_j$.

Lemma 4.1 (Plate 0.5)**.** *Each transverse bivector $\sigma_i \sigma_j$ $(i \neq j)$ squares to $-I_2$.*

Proof. Direct computation, verified symbolically: $(\sigma_1\sigma_2)^2 = \sigma_1\sigma_2\sigma_1\sigma_2 = -\sigma_1^2\sigma_2^2 = -I_2$, and similarly for $(\sigma_2\sigma_3)^2$ and $(\sigma_1\sigma_3)^2$. □

A bivector that squares to -1 is the defining algebraic object of a 2-plane rotation: it generates rotations in its plane via the exponential $\exp(\theta\, e_i e_j)$. This is the geometric content of "rotation in the transverse plane is generated by a square root of -1".

5 The pseudoscalar is the imaginary unit

Theorem 5.1 (Plate 0.6: the geometric origin of i)**.** *Let $\mathcal{I} := e_1 e_2 e_3 \in \mathrm{Cl}(3, 0)$ be the oriented volume element (the pseudoscalar). In the spinor representation,*

$$\sigma_1 \sigma_2 \sigma_3 = i\, I_2, \tag{1}$$

and $\mathcal{I}$ is central in $\mathrm{Cl}(3, 0)$ with $\mathcal{I}^2 = -1$.

Proof. Direct multiplication of the matrices in proposition 3.1 (executed in `sympy`); $\mathcal{I}$ commutes with each σ_i by parity of the number of swaps. □

This is the structural step usually omitted in textbook QM. The imaginary unit appearing in $\hat{p} = -i\hbar\partial_s$ and in the phase $e^{-iEt/\hbar}$ is, in this picture, the spinor shadow of the oriented volume element of the three orthogonal rung directions of GToR. Equation (1) is a theorem, not a postulate.

6 Canonical quantization as a theorem

Theorem 6.1 (Plate 0.7)**.** *A spinor wavefunction propagating along the helix picks up a translation along e_3 and a rotation in the transverse plane. The rotation generator is $J = e_1 e_2$, which satisfies $J^2 = -I_2$, and the exponential*

$$\exp(\theta J) = \cos\theta\, I_2 + \sin\theta\, J \tag{2}$$

defines a one-parameter family of rotations in the (e_1, e_2)-plane. In the spinor representation $J \mapsto i\sigma_3$, so $\exp(i\theta\sigma_3) = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$ is the spin-$\frac{1}{2}$ phase.

The Hermitian generator of translation along the helix is

$$\hat{p} = -\mathcal{I}\,\hbar\,\partial_s \quad \xrightarrow{\text{spinor rep}} \quad \hat{p} = -i\hbar\,\partial_s. \tag{3}$$

Proof. Equation (2) is the bivector form of Euler's identity, which follows from $J^2 = -I_2$ by series expansion. In the spinor representation, $J = \sigma_1\sigma_2 = i\sigma_3$ by direct multiplication; the exponential is then diagonalised in the σ_3 eigenbasis. The translation generator on a helix carries the conjugate momentum; demanding self-adjointness fixes the prefactor up to the universal action quantum, which is the structural role of $\hbar$ (section 11). □

The factor of i in canonical quantization is therefore the spinor image of $\mathcal{I}$. It is not a postulate of the theory: it is a feature of $\mathrm{Cl}(3,0)$ and the helix geometry.

7 The Clifford algebra over Lorentz signature

To pass from $\mathrm{Cl}(3,0)$ to the Lorentz Clifford algebra Cl(1,3) used by the Dirac operator, time is taken as the distinguished slot and the construction proceeds by tensoring with σ_z:

$$\gamma^0 = \sigma_z \otimes I_2, \quad \gamma^i = i\sigma_y \otimes \sigma_i, \quad i = 1, 2, 3.$$

Proposition 7.1 (Plate 2)**.** *With $\eta^{\mu\nu} = \mathrm{diag}(1, -1, -1, -1)$,*

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} I_4 \quad \textit{for all } \mu, \nu \in \{0, 1, 2, 3\}.$$

Proof. By direct symbolic expansion (verified in the notebook): the 10 independent residuals $\{\gamma^\mu, \gamma^\nu\} - 2\eta^{\mu\nu} I_4$ return the zero matrix. □

Remark. The Lorentzian signature itself — which slot of the Clifford algebra gets the minus sign — is *not* derived from the seven GTOR axioms in this paper. We discuss this open hole in section 12.

8 The Dirac operator is forced

The standard textbook route writes the Dirac operator down "as the linearisation of the Klein–Gordon operator" and verifies it works. Here we run that argument as a uniqueness statement.

Theorem 8.1 (Plate 2.5)**.** *Consider the most general first-order ansatz on 4-spinors,*

$$D = \alpha\,\gamma^\mu p_\mu + \beta\, I_4, \qquad \bar{D} = \alpha'\,\gamma^\mu p_\mu + \beta'\, I_4. \tag{4}$$

Demanding $\bar{D}\,D = (p{\cdot}p - m^2)\, I_4$ together with the Clifford relation of proposition 7.1 forces, up to an overall phase,

$$\bar{D}\,D = (p{\cdot}p - m^2) I_4 \iff \beta = -m\alpha,\ \alpha' = \alpha,\ \beta' = m\alpha.$$

Choosing the standard normalisation $\alpha = 1$ gives

$$\hat{D}(p) = \gamma^\mu p_\mu - m\, I_4,$$

the Dirac operator in momentum space.

Proof. Substitute (4) into $\bar{D}D = (p{\cdot}p - m^2)I_4$, expand using proposition 7.1, and split into three scalar equations matching the coefficient structure $(p^2, p, 1)$. The system has a one-parameter family of solutions (overall scale), pinned by $\beta = \pm m\alpha$ and $\alpha' = \alpha$, $\beta' = -\beta$. The calculation is verified symbolically in the notebook. □

Remark (the only structural import here). The first-order restriction on D is itself a structural choice, not proved from the axioms. It is motivated by the fact that differential operators on a manifold are constructed from vector fields (first-order) and connections; higher-order operators are not structurally primitive in the sense of (Lawson and Michelsohn, 1989). We flag this honestly in section 12.

9 From Dirac to Klein–Gordon

Corollary 9.1 (Plates 3–5)**.** *Position-space substitution $p_\mu \mapsto -i\partial_\mu$ in theorem 8.1 gives*

$$\bar{D}\, D = -(\Box + m^2)\, I_4,$$

so each spinor component of a solution of the Dirac equation $D\psi = 0$ satisfies the Klein–Gordon equation $(\Box + m^2)\psi_a = 0$.

Proof. The substitution turns $p_\mu p^\mu$ into $-\Box$; the $\bar{D}D$ expansion of theorem 8.1 then collapses to a multiple of I_4 after the Clifford simplification. □

A spatial Fourier mode reduces the KG equation to an ODE in time:

$$\psi(\boldsymbol{x}, t) = \psi_k(t)\, e^{i\boldsymbol{k}\cdot\boldsymbol{x}} \implies \ddot{\psi}_k + \omega^2\, \psi_k = 0, \qquad \omega^2 := |\boldsymbol{k}|^2 + m^2. \tag{5}$$

10 Riccati reduction to Schrödinger

This is the load-bearing novel piece of the chain. It replaces the WKB ansatz $\psi = A(x,t)\, e^{iS(x,t)/\hbar}$ with an algebraic change of variables on the mode ODE, giving an unambiguous slow-manifold expansion in k/m.

Lemma 10.1 (Plate 7)**.** *Let $u(t) := \dot{\psi}_k/\psi_k$. Then ψ_k solves* (5) *if and only if u solves the Riccati equation*

$$\dot{u} + u^2 + \omega^2 = 0. \tag{6}$$

Proof. Direct: $\ddot{\psi}_k/\psi_k = \dot{u} + u^2$. □

Proposition 10.2 (Plates 8–10: slow-manifold expansion)**.** *The fixed-point set of* (6) *is $\{u = \pm i\omega\}$. The positive-frequency branch $u_{\text{slow}} = -i\omega$ has the convergent expansion*

$$u_{\text{slow}} = -i\sqrt{m^2 + |\boldsymbol{k}|^2} = -im\left(1 + \frac{|\boldsymbol{k}|^2}{2m^2} - \frac{|\boldsymbol{k}|^4}{8m^4} + O(|\boldsymbol{k}|^6)\right). \tag{7}$$

Integrating gives the wavefunction

$$\psi_k(t) = \exp\!\int u_{\text{slow}}\, \mathrm{d}t = e^{-imt}\, \exp\!\left(-\frac{i\,|\boldsymbol{k}|^2}{2m}\, t + O\!\left(\frac{k^4 t}{m^3}\right)\right). \tag{8}$$

The phase e^{-imt} is the leading slow-manifold term, derived rather than imposed.

Theorem 10.3 (Plate 11: Schrödinger from the envelope)**.** *Define the rest-mass envelope* $\phi_k := e^{imt}\,\psi_k$*. Then*

$$i\,\partial_t\phi_k = (\omega - m)\,\phi_k. \tag{9}$$

Truncation of $\omega - m$ *at* $O(k^2/m^2)$ *yields the free Schrödinger equation*

$$i\partial_t\phi_k = \frac{|\boldsymbol{k}|^2}{2m}\,\phi_k,$$

which inverse-Fourier-transforms to

$$i\partial_t\phi(\boldsymbol{x},t) = -\frac{1}{2m}\,\nabla^2\phi(\boldsymbol{x},t).$$

Proof. Differentiate $\phi_k = e^{imt}\psi_k$ and use (8); the Riccati structure forces the envelope ODE. The leading-order coefficient is fixed by the $|\boldsymbol{k}|^2/(2m)$ term in (7). Restoration of $\hbar$ proceeds by dimensional reinstatement, with $\hbar$ playing the role fixed by theorem 11.1 below. □

A complete proof, including bounds on the truncation error, appears in the standalone manuscript (Yazdani, 2026c). The general theory of Riccati reductions is classical (Reid, 1972); the geometric setting for slow-manifold arguments over fiber bundles is (Nakahara). The geometric-algebra framing of sections 3 to 6 follows (Hestenes, 1966; Hestenes and Sobczyk, 1984).

11 Loop integral and the structural role of $\hbar$

The helix is a closed 1-cycle in the transverse plane: winding number one per pitch. This is topology, not choice. A wavefunction ψ on the helix is single-valued after one full turn, $\psi(s+L) = \psi(s)$.

Theorem 11.1 (Plate 13: the action quantum)**.** *Let* $\omega = p_s\,\mathrm{d}s$ *be the canonical* 1*-form on the helix and let* L_{turn} *be the geodesic length of one turn. Single-valuedness of* $\exp(\frac{i}{\hbar}\int\omega)$ *forces*

$$\oint_{\text{turn}} \omega = 2\pi\hbar\,n, \qquad n \in \mathbb{Z}.$$

The smallest non-trivial winding $n = 1$ *defines*

$$\boxed{\hbar = \frac{1}{2\pi}\oint_{\text{turn}} \omega} \tag{10}$$

For a free electron parametrised on the natural Compton turn ($L_{\text{turn}} = \lambda_C$, $v = c$, $m = m_e$)*,* (10) *gives* $\hbar = m_e\,c\,\lambda_C/(2\pi)$*.*

Proof. The Bohr–Sommerfeld quantisation rule (Bohr, 1913; Sommerfeld, 1916; Arnold, 1989): requiring single-valuedness of the canonical phase factor on a closed cycle quantises the holonomy of the 1-form. Dimensionally one verifies $[\,m\,v\,L\,] = M\cdot LT^{-1}\cdot L = ML^2T^{-1} = [\text{action}]$ ✓. The pure rotation limit $h_p \to 0$ recovers the Bohr angular-momentum quantum $\hbar = R\,m\,v$. □

What "deriving $\hbar$" means in post-2019 SI. Bureau International des Poids et Mesures (2019) fix $\hbar = 1.054571817\times10^{-34}$ J s, e, and c by definition. Consequently $\hbar$ cannot be *derived* in the sense of an empirical prediction; it is a unit convention. What (10) delivers is the structural *role* of $\hbar$ as the action per natural turn divided by 2π. The numerical match $\hbar_{\text{pred}} = m_e c\lambda_C/(2\pi) = \hbar_{\text{SI}}$ to 6×10^{-10} relative is a tautological unit-bridge, since $\lambda_C := h/(m_e c)$ is itself defined in terms of h.

The genuine geometric prediction is α. The dimensionless geometric output of GToR is the fine-structure constant. From Cor 30, $14\pi^2 = \sqrt{2}\, S\, V$, with S the boundary surface and $V = 7\pi/3$ (Cor 27); from Cor 31,

$$\alpha_\infty = \frac{1}{14\pi^2} = 7.2372274 \times 10^{-3}, \tag{11}$$

to be compared with $\alpha_{\text{CODATA}} = 7.2973526 \times 10^{-3}$ (Tiesinga et al., 2021). The relative deviation is -8.24×10^{-3}. The atlas correction (Cor 34) closes this to five significant figures (Yazdani, 2026b).

The SI relation $\alpha = e^2/(4\pi\varepsilon_0\hbar c)$ shows how (11) interfaces with the SI: with exact $e, c, \hbar$ as in the post-2019 system, a geometric value of α *determines* ε_0. In the pre-2019 SI it would have determined $\hbar$. Either way, the dimensionless content lives in α, not in the numerical value of $\hbar$.

12 Scope, structural choices, and open holes

We list the imports of this chain, each with the honest status of its provenance.

Hole 1: the Lorentzian signature. We work with $\eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$ from section 7 onward. $\text{Cl}(3, 0)$ is Riemannian, and the passage to Cl(1,3) via $\otimes\sigma_z$ is the standard construction (Cartan, 1938; Lawson and Michelsohn, 1989); but *which* slot becomes the distinguished time slot is not derived from the seven GToR axioms in this paper. The candidate derivation uses Cor 23 (geometric expansion of space) for the distinguished slot and Cor 33–38 (self-similar atlas, even parity of R-corrections) for the $-$ sign role; that derivation is not executed in the present notebook.

Hole 2: "$\hbar$ from geometry". Reframed and closed at the honest level. Within post-2019 SI, $\hbar$ is definitional; (10) is the structural statement, and the empirical content sits in α.

Other acknowledged imports.

(i) *First-order operator restriction (theorem 8.1).* Motivated by structural primitivity in the sense of (Lawson and Michelsohn, 1989); not proven from GToR axioms.

(ii) *Positive-frequency branch (proposition 10.2).* The Riccati fixed-point set contains both $\pm i\omega$; choosing $-i\omega$ selects the physical sheet. Standard in QM; not proven here.

(iii) *Rest-mass envelope subtraction (theorem 10.3).* Defining $\phi := e^{imt}\psi$ is the standard non-relativistic reduction move; it is a change of variable, not a derivation.

What this paper does and does not claim. The paper is *not* a re-derivation of quantum mechanics. Its contribution is two-fold. First, structural: relocating three pieces of QM machinery (the imaginary unit, canonical quantization, the Dirac operator) inside the geometric setting of GToR, where they are theorems rather than postulates. Second, technical: replacing the WKB ansatz in the Klein–Gordon $\to$ Schrödinger reduction with an algebraic Riccati slow-manifold expansion (section 10, (Yazdani, 2026c)). The dimensionless prediction $\alpha_\infty = 1/(14\pi^2)$ is inherited from the parent manuscript (Yazdani, 2026b) and used here only as the natural endpoint of the chain.

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A Notebook artifact

The companion artefact for this paper is the Jupyter notebook `framable_dirac_to_schrodinger.ipynb` (v2, 39 cells, 18 executable). Each numbered Plate in the notebook corresponds to a section in the paper as follows:

Notebook Plate	**Paper section**
Plate 0	section 3
Plate 0.5	section 4
Plate 0.6	section 5
Plate 0.7	section 6
Plates 1–2	section 7
Plate 2.5	section 8
Plates 3–5	section 9
Plates 6–11	section 10
Plate 13	section 11
Open Holes	section 12

Every code cell in the notebook is a sympy verification of the algebraic claim in the corresponding plate. Outputs are LaTeX displays.